Homework 4

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Mean Model Selection

1. Candidate mean structures (15 points): Fit at least three candidate mean models to your data. These might include a saturated (profile) model, polynomial models (linear, quadratic), or spline models. For each model, report the number of parameters and the log-likelihood.

As established in the previous assignment, we will use AR(1) as our working covariance model to compare mean models. First, we fit a saturated model, where the time period is treated as a categorical variable.

```
##                 Estimate Naive S.E.      Naive z Robust S.E.     Robust z
## (Intercept) 129.023110785  0.5835132 221.11428158   0.4588785 281.17050287
## PERIOD2       5.551724138  0.4558768  12.17812434   0.4135381  13.42493903
## PERIOD3      10.250917095  0.5934834  17.27245732   0.5091190  20.13461993
## SEX           1.923715042  0.7696084   2.49960241   0.6888531   2.79263473
## PERIOD2:SEX  -0.009944431  0.6012659  -0.01653916   0.5571205  -0.01784969
## PERIOD3:SEX  -0.339631148  0.7827583  -0.43389019   0.6757889  -0.50256986
```

Next, we fit a linear model, where time is treated as a continuous variable.

```
##                Estimate Naive S.E.    Naive z Robust S.E.    Robust z
## (Intercept) 129.0795727 0.58169343 221.903097  0.45525227 283.5341670
## YEAR          0.8542431 0.04946735  17.268826  0.04242658  20.1346199
## SEX           1.9448911 0.76720822   2.535024  0.68417849   2.8426663
## YEAR:SEX     -0.0283026 0.06524357  -0.433799  0.05631574  -0.5025699
```

Finally, we fit a linear spline model, with a knot at year 6.

```
##                 Estimate Naive S.E.      Naive z Robust S.E.     Robust z
## (Intercept) 129.023110785 0.58351324 221.11428158  0.45887854 281.17050287
## YEAR          0.925287356 0.07597946  12.17812434  0.06892302  13.42493903
## SPLINE       -0.142088530 0.11535838  -1.23171402  0.11614040  -1.22342033
## SEX           1.923715042 0.76960841   2.49960241  0.68885308   2.79263473
## YEAR:SEX     -0.001657405 0.10021098  -0.01653916  0.09285342  -0.01784969
## SPLINE:SEX   -0.053290381 0.15214869  -0.35025198  0.16262494  -0.32768885
```

Each model estimates a scale parameter and a correlation parameter that jointly define the AR(1) covariance structure. In addition to this, the saturated mean model consists of 6 parameters, the linear mean model consists of 4 parameters, and the spline mean model consists of 6 parameters.

These GEE models make no distributional assumptions about the outcome, so there are no log-likelihoods to report.

2. Model comparison (15 points): Compare your candidate mean models using appropriate criteria. Select a mean structure and justify your choice. If you conduct any hypothesis tests to compare nested models, explain your approach.

The linear model is nested in the spline model, so we can compare them using a Wald test with H_0 : $\beta_{SPLINE} = \beta_{SPLINE:SEX} = 0$. We can also write this as $Q'\beta = \mathbf{0}$ where $Q' = \left(\begin{smallmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{smallmatrix}\right)$ and $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4, \beta_5)^T$.

$T_W = (\boldsymbol{Q'}\widehat{\boldsymbol{\beta}})'[\boldsymbol{Q'}\widehat{Var}(\widehat{\boldsymbol{\beta}})Q]^{-1}(\boldsymbol{Q'}\widehat{\boldsymbol{\beta}}) \sim \chi^2_2$ under H_0.

After running the test with 2 degrees of freedom, we get a test statistic of 4.44 and a p-value of 0.108. Since the p-value is > 0.05, we fail to reject the null hypothesis and conclude that the spline model does not explain significantly more of the variation in SBP over time, when compared to the linear model.

```
##   Statistic DF   P_value
## 1   4.44259  2 0.1084686
```

From here on, we compare the linear model to the saturated model. Since the GEE model is not likelihood-based and the two remaining models are non-nested, we will compare their goodness of fit using QIC.

```
QIC(fit_sat, fit_linear)
```

```
##                 QIC
## fit_sat    9623.663
## fit_linear 9624.394
```

The saturated model has a lower QIC than the linear model. Therefore, we will choose the saturated model as our final mean model.

Covariance Model Selection

1. Candidate covariance structures (15 points): Using your selected mean structure, fit your model with at least three different covariance structures. For each covariance structure, report the number of covariance parameters and the log-likelihood.

As noted before, we will use the saturated model as our mean structure. The three covariance structures that we will test are AR(1) (used in mean model selection), unstructured, and compound symmetry. Both the AR(1) and compound symmetry models estimate 2 covariance parameters, σ^2 and ρ. The unstructured covariance model estimates $n(n+1)/2 = 3(4)/2 = 6$ covariance parameters. The table below gives the AIC, BIC, and REML log-likelihood that each model yields.

2. Model comparison (15 points): Compare your candidate covariance structures using appropriate criteria. Select a covariance structure and justify your choice. If any comparisons involve testing whether a variance component equals zero, discuss how the boundary issue affects your approach.

Compound Symmetry and AR(1) are both nested within Unstructured covariance so we can compare these with REML LRT. To compare CS and AR(1) we need to use AIC since these are not nested. When comparing compound symmetry to unstructured, the REML LRT yields a test statistic of 226.98 and a p-value of $< .0001$. The REML LRT for comparing AR(1) to unstructured yields a test statistics of 340.04 and a p-value of $< .0001$. These tests tell us that in both cases, unstructured covariance is preferred over the simpler structures. For completeness, we found that AIC for CS is 82027.30 and for AR(1) the AIC is 82140.36, which leads to us choose compound symmetry over AR(1) since it has a smaller AIC. Based on all these criteria, we will choose unstructured covariance for our final model. Although it requires estimating more parameters, we found statistically significant evidence that estimating more parameters improves the fit of our model.

Note that using REML LRT is not valid with when we are testing that a variance component equals zero. This is because variances are bounded below by zero, and when we test if a variance equals 0, this places the null hypothesis on the boundary of the parameter space. The LRT is only valid when the null value is on the interior of the parameter space, so this test is not valid in cases when a variance parameter equals 0. This does not arise in our case.

```
##        Model df      AIC      BIC    logLik   Test  L.Ratio p-value
## fit_cs     1  8 82027.30 82084.66 -41005.65                        
## fit_un     2 12 81808.31 81894.36 -40892.16 1 vs 2 226.9823  <.0001
```

```
##         Model df      AIC      BIC    logLik   Test  L.Ratio p-value
## fit_ar1     1  8 82140.36 82197.72 -41062.18                        
## fit_un      2 12 81808.31 81894.36 -40892.16 1 vs 2 340.0427  <.0001
```

Model Comparison
Covariance Structure Selection

Covariance Structure	# Cov Params	AIC	BIC	Log-Likelihood
Compound Symmetry	2	82,027.30	82,084.66	-41,005.65
AR(1)	2	82,140.36	82,197.72	-41,062.18
Unstructured	6	81,808.31	81,894.36	-40,892.16

Final Model

1. Model specification (10 points): State your final model, including the mean structure and covariance structure. Explain why you chose these structures.

Our final model uses the saturated model as the mean model and unstructured covariance for the covariance structure. We chose the saturated model since it has a lower QIC than the linear model. We chose unstructured covariance since we found that it fits the data better than both compound symmetry and AR(1) when compared using REML LRT.

2. Parameter estimates (15 points): Report the fixed effect estimates with standard errors and 95% confidence intervals. Report the covariance parameter estimates. Interpret the key parameters in the context of your research question.

Find all estimates, standard errors, 95% confidence intervals, and p-values in the table below. The estimated covariance matrix is provided above the table. $\widehat{\beta_0} = 129.02$ represents the mean SBP for males at the first visit. $\widehat{\beta_1} = 5.55$ and $\widehat{\beta_2} = 10.25$ represent the change in SBP from baseline to visits 2 and 3 respectively. $\widehat{\beta_4} = 1.92$ represents the difference in SBP between males and females at baseline. All of these terms are significant, indicating significant differences in SBP both over time and between males and females (evaluated separately). $\widehat{\beta_5} = -0.01$ and $\widehat{\beta_6} = -0.34$ represents the difference in SBP between males and females at later visits. Neither of these terms is significant, indicating that the difference in SBP between males and females does not change over time.

```
##          [,1]     [,2]     [,3]
## [1,] 401.9169 310.2227 283.5815
## [2,] 310.2227 465.4358 334.6832
## [3,] 283.5815 334.6832 524.9023
```

Final Model

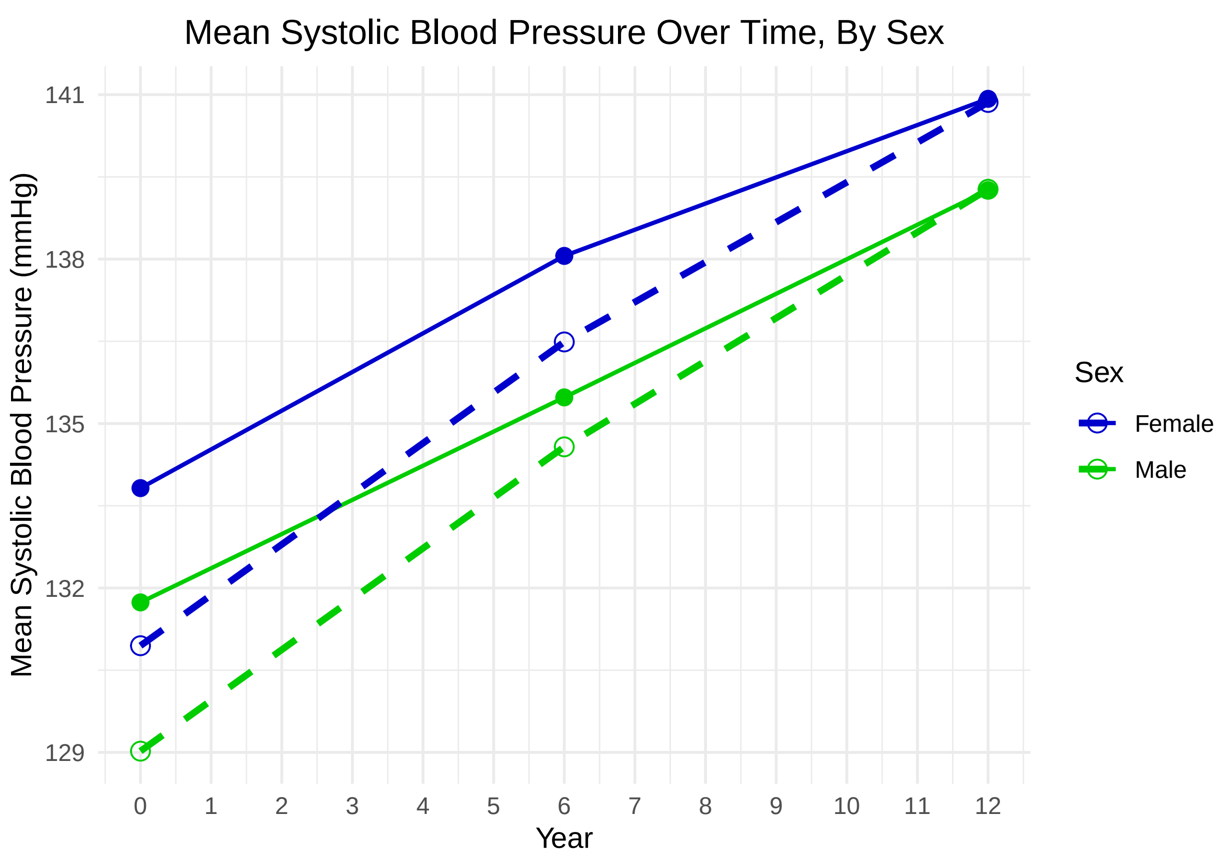
Saturated Mean Model, Unstructured Covariance

Term	Estimate	Standard Error	P-Value	CI-Lower	CI-Upper
(Intercept)	129.02	0.54	0.00	127.96	130.09
PERIOD2	5.55	0.43	0.00	4.72	6.39
PERIOD3	10.25	0.51	0.00	9.24	11.26
SEX	1.92	0.72	0.01	0.52	3.33
PERIOD2:SEX	-0.01	0.56	0.99	-1.11	1.09
PERIOD3:SEX	-0.34	0.68	0.62	-1.67	0.99

Visualization and Interpretation

1. Visualization (10 points): Create a figure showing the observed data (e.g., group means over time) overlaid with the fitted trajectories from your final model.

The observed data trajectories are plotted with solid lines, and the fitted trajectories from the final model are plotted using dashed lines.



2. Summary (5 points): In 1–2 paragraphs, summarize your model-building process and key findings as you would for a collaborator.

Our model-building process was conducted in two stages. In the first stage, we used GEE models to compare mean model structures using Wald tests and QIC. At the end of the comparison process, we found that a saturated model structure was the best fit for our data. In the second stage, we used GLS models to compare covariance model structures using REML LRT and AIC. At the end of the comparison process, we found that an unstructured covariance model was the best fit for our data. Our final model is a GLS model that combines the saturated mean model and the unstructured covariance model to fit population mean SBP trajectories for both males and females.

We found that when sex is fixed, mean SPB increases by about 5.55 mm/Hg after 6 years past baseline, and it increases by about 10.25 mm/Hg after 12 years past baseline. Both of these time effects were found to be statistically significant, which suggests that on average, systolic blood pressure worsens with age for both males and females. We also found a statistically significant difference in SBP between males and females at each time point: an increase of about 1.92 mm/Hg for females when compared to males. This suggests that on average, females have higher systolic blood pressure than males at any given time period. Lastly, the interaction effect between sex and time period was not found to be statistically significant, which suggests that the rate of change of SBP over time does not differ between the sexes.